SQL and DBMS

1. Consider the following relation: Transactions(TransactionID, AccountID, BranchName, Amount, TransactionDate) Write an SQL query to display the branch name, average transaction amount, and total number of transactions for each branch where the average transaction amount exceeds BDT 30,000 and the total number of transactions in that branch is at least 10.

2. You have tables: EMPLOYEES (ID, Name, DeptID) and DEPARTMENTS (DeptID, DeptName). Write the logical steps (or pseudo-SQL logic) to get a list of employee names who work in the "IT" department.

3. STUDENTS (StudentID, Name, Department, Marks)
I. Write an SQL query to display only StudentID, Name, and Marks for students scoring more than 75 marks.
II. Write an SQL query to count how many students scored more than 75 marks in each department.

4. Hospital (HospitalID, HospitalName, Specialization, PatientCount) Write an SQL query to display the names of the five hospitals that specialize in cardiology and have treated the highest number of patients.

5. A developer writes the SQL statement SELECT EmployeeName, Salary FROM Employees WHERE Salary > AVG(Salary); but the query generates an error. Explain why the query generates an error and rewrite it correctly to display employees whose salaries are greater than the overall average salary.

6. Write an SQL query to retrieve the Department, the total number of employees, the total salary expenditure, and the average salary for each department. The output should display one record for each department. Use the table Employees (EmpID, EmpName, Department, Salary).

7. An institute wants to create a database table named STUDENT to store student information. The table should include the columns Roll Number, Name, Department, Email, and Admission Date. Specify the most appropriate SQL data type for each column and identify which column should be defined as the Primary Key, giving a brief justification for your choice.

8. Employees (EmpID, Name, Department, Salary) Write an SQL query to retrieve employees whose salaries are greater than the average salary of their respective departments.

9. Write an SQL query to find duplicate names in the employee table Employee (EmpID, EmpName, Department, Salary). Display each duplicate name along with the number of times it appears.

10. Employee(E_ID, Ename, M_ID, Dept_ID, Salary) Department(D_ID, Dept_Name)

- Find out the names of the manager for each employee.
- Calculate the total salary of each department and sort the departments based on total salary in descending order.

11. Catalog (sid, pid, price) Supplier (sid, sname, address) Product (pid, pname, cost) Find the supplier name (sname) who supplies the product with pname = 'Monitor' at the minimum price.

12. Let a database have two tables, Customers (CustomerID, CustomerName) and Orders (OrderID, CustomerID, OrderAmount). Explain INNER JOIN, LEFT JOIN, RIGHT JOIN, and FULL OUTER JOIN. Write the SQL queries and show the result sets of each join operation using suitable sample data.

13. Employee (ID, Name, Salary) Equipment (ID, Name, Price) Hire (EmployeeID, EquipmentID) Write an SQL query to show the name of the employee who hired the maximum number of distinct equipment items. If multiple employees have the same maximum count, display all of them.

14. Given the following two tables, Students (StudentID, StudentName) and Marks (StudentID, SubjectID, Mark), write down the output of the given SQL queries and write SQL queries for the following outputs:

I. SELECT COUNT(*) FROM Students;

II. Display the names of students whose total marks are at least 250.

III. List all students' names and the number of subjects in which they have obtained marks.

IV. List all students who have not obtained marks in any subject.

V. List all distinct subject IDs from the Marks table.

15. A database has a Managers table and an Employees table. The Employees table references the Managers table using a Foreign Key. The database uses different deletion rules.

I. Explain what ON DELETE CASCADE means in simple English. If one manager is deleted, what happens to the employees under that manager?

II. Explain what ON DELETE RESTRICT means. What happens if someone tries to delete a manager who still has employees?

III. Explain what happens when a Foreign Key uses ON DELETE SET NULL. What condition must be satisfied by the Foreign Key column?

IV. Compare ON DELETE CASCADE, ON DELETE RESTRICT, and ON DELETE SET NULL. Which approach is safer for a banking system, and why?